Andhra University College of Engineering

Chemical Engineering | Course Code: CH301

Chemical Reaction Engineering - Laboratory Manual & Practical Guide

Syllabus Focus: CH301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Kinetics of Homogeneous Reactions

" Syllabus Scope: Rate of reaction, Temperature dependency, Arrhenius law, Collision theory, Transition s

Unit 2: Ideal Reactor Design Equations

" Syllabus Scope: Design equations for Batch, Continuous Stirred Tank CSTR, and Plug Flow Reactors PFR un

CH301: Chemical Reaction Engineering - Advanced Topics & Assessment

Academic Track: Chemical Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Multiple Reactor Networks & Yield

" CSTRs in series and parallel, Comparison of reactor volumes for first and second order reactions, Product se

Unit 4: Non-Isothermal Reactor Design

" Energy balance for batch, CSTR, and PFR, Adiabatic operations, Equilibrium conversion, Optimum temperature p

Unit 5: Non-Ideal Flow & Residence Time Distribution

" Residence time distribution RTD, Et and Ft curves, Pulse and step tracer stimulus experiments, Tank-in-serie

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Kinetic Study of Saponification of Ethyl Acetate in an Isothermal Batch Reactor

Lab 2: Performance Evaluation of a Continuous Stirred Tank Reactor CSTR at Steady State

Lab 3: Residence Time Distribution RTD Measurement in a Plug Flow Reactor using Pulse Tracer

Lab 4: Comparison of Conversion in a Cascade of Three CSTRs in Series vs Single PFR

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* H. Scott Fogler - Elements of Chemical Reaction Engineering Prentice Hall 5th Ed

* Octave Levenspiel - Chemical Reaction Engineering Wiley 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students